How-To Library

Informatica

# Informatica MDM - Customer 360 SaaS Best Practices Guide

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# Abstract

This article contains information about the best practices to configure MDM SaaS for managing customer records. These best practices have been derived from real-world experiences that cover different use cases.

# Supported Versions

- Informatica MDM - Customer 360 SaaS July 2026

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# Overview

MDM SaaS is a cloud-based master data management platform that you can use to manage records. You can configure the match and merge process to identify and resolve duplicate records. To identify duplicates, you can choose to use a machine learning (ML) model, declarative match rules, or both.

# Prerequisites

Before the Informatica team can provision the MDM SaaS environments, ensure that your IT administrator completes the required prerequisite questionnaires and checklist.

# Key terminologies

The following table lists the terms that Informatica Intelligent Data Management Cloud uses:

| Term | Description |
|---|---|
| Source | Location from where you retrieve data. |
| Target | Location to which you move the data. |
| Connection | Provides access to data in cloud and on-premise applications, platforms, databases, and flat files. |
| Org | A secure area within the Informatica Intelligent Data Management Cloud repository that stores your licenses, user accounts, assets, such as business entities and reference data. You might have access to one or more organizations. |
| Secure Agent | A lightweight program that runs all tasks and enables secure communication across the firewall between your organization and Informatica Intelligent Cloud Services. |
| Asset | An object that can be created, read, updated, and deleted in Informatica Intelligent Data Management Cloud. |

# Secure Agent

A Secure Agent is a lightweight program that runs all tasks and enables secure communication across the firewall between your organization and Informatica Intelligent Data Management Cloud.

For more information about getting started with Secure Agents, see the Minimum requirements and best practices when installing Informatica Cloud Secure Agent Knowledge article.

# Provisioning an organization

Before provisioning an organization, consider the following guidelines:

- Before you set up Customer 360 SaaS, upgrade the organization to the latest release from Business 360 Console.
- Set up Customer 360 SaaS before migrating any assets.
- Assign the Admin role and a group to users.
- Create projects for additional assets that you require, such as mappings, mapping tasks, taskflows, and other MDM SaaS assets. Avoid using predefined projects, such as Business360, Reference360, and Customer360, for these assets.
- Don't add any permissions to the predefined projects on the Explorer page in MDM SaaS.

# Adding IP addresses to approved list

If you use protective firewall, then add the IP addresses that are specific to your POD to the firewall allow list.

For information about the IP addresses that are specific to your POD, see POD Availability and Networking. For information about the IP addresses that are specific to MDM SaaS, see the IP addresses for MDM SaaS Knowledge article.

After you add the IP addresses to the firewall allow list, you can validate whether you have successfully added them. For more information about validating the IP addresses that you added to the firewall allow list, see the Validate IP addresses for MDM SaaS Knowledge article.

**Note:** Informatica doesn't add any source specific IP addresses to allow list. After the July 2023 release, any firewall rules that open a MongoDB connection for the Secure Agent become obsolete. When you use a proxy configuration, a Business 360 FEP connection fails. To fix this issue, see the Business 360 FEP Test Connection Failure Knowledge article.

# Designing data model

Before you design the data model, ensure that you consider the following guidelines:

- Identify the source systems from which you load data into MDM SaaS. Note that MDM SaaS isn't based on a relational data model and instead uses a document-type approach or hierarchical model.
- To map source system attributes with MDM SaaS attributes, use the Customer 360 SaaS data dictionary. For more information, see the Customer 360 SaaS data model Knowledge article and Customer 360 SaaS data dictionary.
- After you finalize the data model, create other assets for your business case.
- Don't delete or remove any predefined assets. For more information about customizing predefined assets, see Customize predefined assets.
- When you create custom assets, use the following naming convention that differentiates predefined and custom assets:
  `X_<asset name>`

# Ingress job

Before you create an ingress job, consider the following guidelines:

- Identify the differences between the Business 360 connector and the Business 360 FEP connector. For more information about the differences between the Business 360 and Business 360 FEP connectors, see the FAQ: Difference between Business 360 connector & Business 360 FEP connector Knowledge article.
- Consider the data source for the ingress jobs because not all the connectors available in Informatica Intelligent Cloud Services can be used in the mapping for ingress jobs. For example, you can't use the Databricks Delta connector with custom queries in an ingress mapping for MDM SaaS. Before you define an ingress job, identify the source systems and validate the connectivity.
- Validate the connectivity with the MongoDB Atlas instance and the Business 360 connector.
- Use Business 360 Flat End Point (FEP) connector instead of Business 360 connector to ensure parallel execution and performance.
- To load a multibyte data set, meet the following requirements:
  - Define JVM options in the Secure Agent.
    For more information, see Rules and Guidelines for Business 360 FEP Connector.

    For more information about loading a multibyte data set with ingress, see the Multibyte characters not loading properly with Ingress when using Business 360 connector Knowledge article.
  - The source primary keys of records can contain the following characters:

```
~!'={}|:@#$^&*()-_+,<>?`
```

The source primary keys of records don't contain a slash (/), backslash (\), period (.), percentage (%), semicolon (;), or double quotation mark (").

**Note:** You can use special characters in source primary keys when you add records through ingress jobs or file import. However, if you edit a record that contains any special characters that aren't allowed in its source primary key on the record details page, MDM SaaS fails to update the record.

- The source primary keys can contain up to 255 special and alphanumeric characters, with English letters only. If the source primary keys contains more than 255 characters, the additional characters are truncated.
- The source primary keys of field groups can contain the following special characters:

  colon (:), hyphen (-), pipe (|), period (.), or underscore (_).

  **Note:** You can use special characters in source primary keys when you add field group entries through ingress jobs or file import. However, if a field group entry contains special characters that aren't allowed in its source primary key, you might not be able to create, edit, and view field group values on the record details page.

**Note:** When you schedule ingress jobs to run on an hourly basis, the jobs fail due to previous running jobs or huge ingress data. To avoid job failures, don't schedule ingress jobs on an hourly basis. Additionally, you can run ingress jobs using the Create Ingress Job Instance API to import data from an external source system into the Business 360 data store.

## Egress job

Before you create an egress job, consider the following guidelines:

- To export all records successfully, create or update records before you run an egress job.
- Add a sorter transformation before all joiner and aggregator transformations.
- When you design the data model, keep field names short because long field names increase the overall size of the JSON file.
  For example, instead of AlternateIdentifier field group name and the AlternateIdentifierValue and AlternatiIdentifierType field names, use AltId, value, and type field names.

  If you have 1000 alternative identifiers, it saves 40,000 characters.
- To reduce the size of the read buffer, remove auto-mapping and select only the columns that you need.
- Review and modify the disk space in the Secure Agent file system.
- Cache Size for Web Service Response (KB) can be changed for source and web service. The maximum cache size supported is 100 MB. A larger cache size impacts the buffer size calculation.
- Review higher precision. Default is 255 for string. You can decrease it based on your data model.
- Increase buffer pool size if the following warning appears:

  ```
  TM_6351 WARNING: Insufficient number of data blocks for adequate performance
  DBG_21699 Waiting for consumer to free buffer block.
  ```
- Increasing DTM buffer size increases buffer blocks.
  To calculate the optimal block size, use the following equation:

  ```
  DTM buffer block size = 2*X where X is the maximum row size of any transformation.

  DTM buffer size = (DTM buffer block size) * 10 * N where N is the number of partitions.
  ```
- Tune sorter cache size for the sorter transformation.
- Tune aggregator data and index cache size for the aggregator transformation.
- Tune joiner data and index cache size for the joiner transformation.

- Tune lookup data and index cache size for the lookup transformation.

**Note:** When you schedule egress jobs to run on an hourly basis, the jobs fail due to previous running jobs or huge egress data. To avoid job failures, don't schedule egress jobs on an hourly basis. Additionally, you can run egress jobs using the Create Egress Job Instance API to export data from the Business 360 data store to an external data source.

## Migrating assets

You can migrate the following assets from one organization to another organization:

- Predefined and custom business entities. A business entity asset includes attributes, data quality rule associations, match model, survivorship configuration, and search configuration.
- Business application
- Pages
- Components
- Job definitions
- Source system
- Business events
- Reference data set and crosswalk configuration
- Hierarchies
- Relationships
- Authorization. The asset contains the privileges configured for different user roles.

You can't migrate the following objects from one organization to another organization:

- Process
- Code values
- Job schedules

Before you migrate assets, consider the following guidelines.

- The source and target organizations must be of the same version.
- Before you import assets into a target organization, export all the assets in the target organization as a backup so that you can import them back in case of any failure.
- Don't make changes in Reference 360, such as renaming or deleting reference data assets, that might create inconsistency in Customer 360.
- A user with the Admin role must migrate the Authorization asset.
- When a user with the Admin role migrates assets to a target organization for the first time, the user can migrate all the assets including the authorization and business event assets.
- When a user with the Designer and MDM Designer roles migrates assets to a target organization for the first time, the user can migrate all the assets except authorization and business event assets. A user with the Admin role can then migrate the authorization asset first and the business event assets later.
- Incrementally, you can migrate the assets that are updated.
- Ensure that you create connections in the target organization before you migrate taskflows to it.
- Before migrating assets to a target organization, don't set any permissions to assets, folders, or projects.
- Use users with the same roles to make changes in the source organization and to migrate them.

For example, if a user with the Admin role updates an asset, ensure that a user with the Admin role migrates it to a target organization. If a user with a different role migrates it, you might get metadata errors.

- During migration, the licenses aren't checked, so ensure that you migrate assets that are licensed in the target organization.

  **Note:** Asset migration uses internal IDs for MDM SaaS assets and names for Reference 360 assets.

- Don't migrate assets without provisioning the target organization. The organization might get corrupted and can't be used.

**Note:** Attribute dependency being used in match model must be manually removed before migrating assets.

## Configuring data quality rule associations

Before you configure data quality rule associations, consider the following guidelines:

- Ensure that production and non-production keys are specified according to usage. The system enforces limited transactions and consumes based on usage.
- Use basic rule associations for business entities involved in high concurrent cleansing transactions.
- Use lighter advanced data quality rule associations.
- Enable DTM caching in mappings so that DTM doesn't initialize mappings every time.
- Don't modify the predefined verifier asset named, Address_Verifier_OOTB.
- Don't rename license file name in the Batch Address Verification section of Global Settings.

## Tasks and workflows

For system-triggered events, you can have a maximum of 100 review tasks. If you use REST APIs to merge records that have an approval workflow associated with them, the tasks will no longer be required. Instead of deleting the tasks from Cloud Application Integration, take any action on the tasks and close them.

When we delete tasks from Application Integration, the tasks get deleted, but the change list IDs associated with these tasks still remain and don't allow you to create any more tasks after the limit of 100 is reached.

## MDM SaaS REST APIs

Before you use MDM SaaS REST APIs, consider the following guidelines:

- Don't run the REST APIs in loops to perform batch processing.
- To process a large volume of data efficiently, use batch jobs instead of REST APIs.
- Run batch loads during off-hours to avoid impacting the performance of the services.
- Specify the model version in each request to avoid server side repeatedly looking for metadata.
- The search match API performance depends on the configured match model, match rules, and data.
- The search match API uses a lot of resource. Use the search API to perform a regular search.
- When you use the job instance APIs, monitor the jobs to ensure that multiple jobs aren't triggered.
- The data quality rule associations and their complexity have a significant performance impact on the business entity create and update APIs.

## Hierarchy and relationships

Before you create hierarchies and relationships, consider the following guidelines:

- Use the Business 360 FEP connector to ingress data.
- Create a separate mapping to create a hierarchy instance, root record, and relationship. Add all the mapping tasks in one taskflow.
- Relationship primary keys between parent and child records across the hierarchy must be unique.

For more information about importing hierarchy data using the Business 360 FEP connector, see Ingress Hierarchy data using FEP connector in Customer 360 SaaS Knowledge article.

**Note:** You can't ingress relationship records by business ID for parent and child records. You can instead use source primary keys.

## Reports

Before you create custom reports, consider the following guidelines:

- To create reports, users require the following user roles:
  - Admin
  - Customer 360 Data Steward
  - Customer 360 Manager
  - MDM Business User
  - Customer 360 Analyst
  - Designer
- When users directly save a report without configuring a chart, they can't add the report to the dashboard. The reports are added to the reports table.
- Only users with the Admin user role can schedule the publishing of reports.
- Users can add one dimension, maximum of five measures, and as many dashboards as they want.

## Defining jobs

To define jobs, use Business 360 Console or the job instance REST APIs.

## Guidelines to improve performance

Performance is a joint responsibility, so the right architecture and design are necessary to build a highly performing application.

Consider the following guidelines:

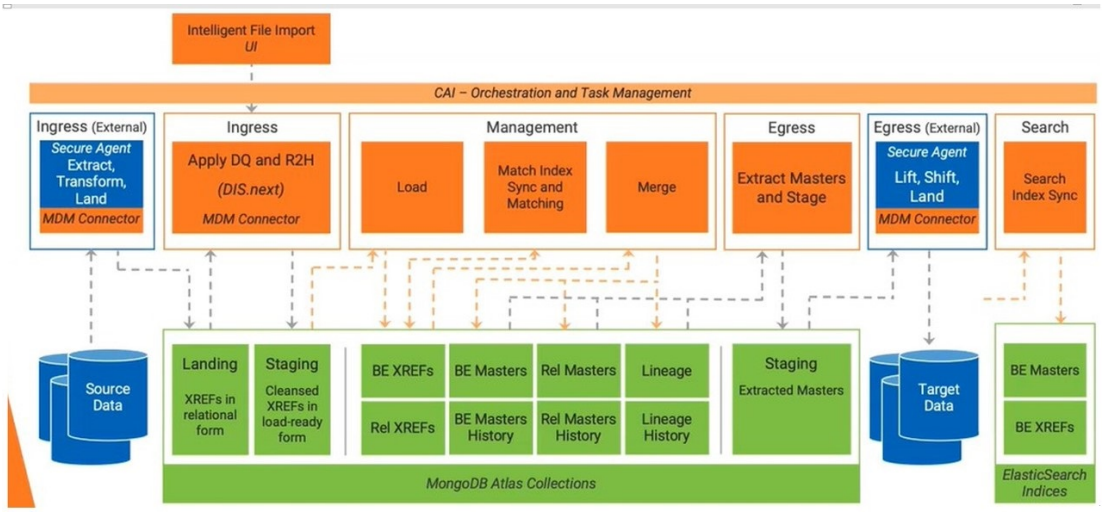
- Perform functional testing on small volumes and ensure that the configuration meets functionality. When you use high volumes, it takes longer time to complete the tests and adds huge load to the Informatica environments. Similarly, for any performance tests, perform incremental tests so that issues are discovered faster and are much easier to replicate.
- Before configuring match rules, perform proper data profiling. Optimal match rules are derived after iterative tests.

- For extra large volumes, use the chunking strategy to load a maximum of 25 - 50 million records as part of one ingress job. To optimize the load time and help with the recovery time in case of any job failures, logically divide extra large source data into multiple jobs.
- Secure Agent is hosted on customer's infrastructure. Work with Informatica to properly size it as it has a considerable impact on preprocessing MDM data.
- The MDM test was done with that partition purely for internal purposes. Customers can choose the number that works best and generally arrive at it after iterative tests. For more information about partitioning a mapping, see Partitions documentation.

## MDM SaaS data flow

When you import a record in to the MDM - Customer 360 SaaS, the record goes through different processes before getting added or merged to the master data.

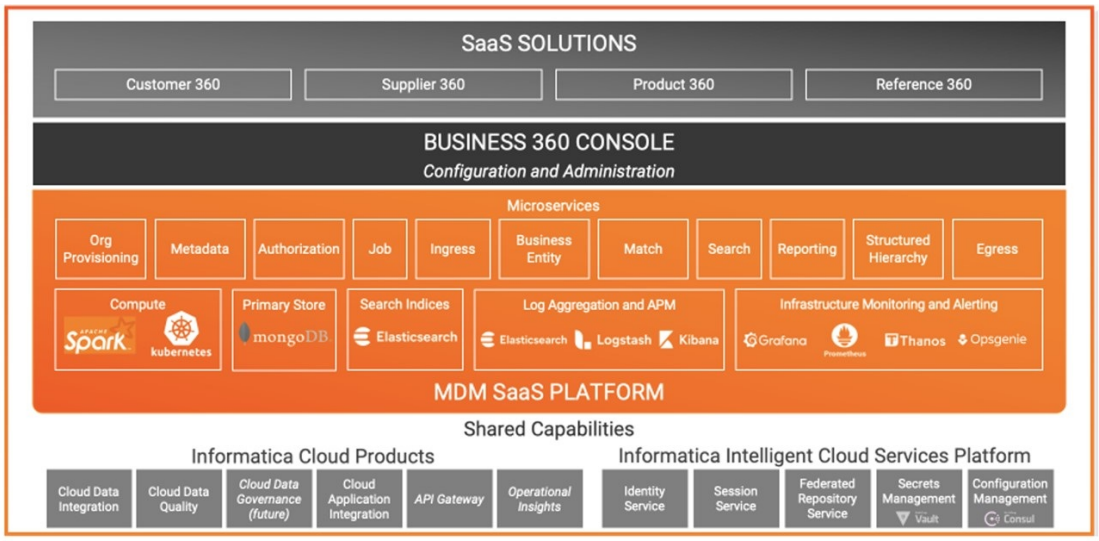
The following image shows the end-to-end flow of a record in MDM - Customer 360 SaaS:




## MDM SaaS components

MDM SaaS uses different components and a host of services for its proper functioning.

The following image shows the different components in MDM SaaS:




# MDM SaaS connectors

MDM SaaS persists data in MongoDB, and there are no direct connections to the database. All data access is through the supported Cloud Data Integration connectors.

The following predefined connectors are available for data access:

- Business 360 connector
- Business 360 FEP connector
- Business 360 Events connector

To ingress data, use Business 360 FEP connector as it reduces the computational requirements on Secure Agent and has huge performance improvements.

To egress all data attributes, use Business 360 connector.

# Author

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